

Queries

1. Select all columns form Employee Table

*select * from Employee*

Database: MySQL v5.7	Run	Update	Fork	Load Example	Star PRO	Embed PRO	Collaborate	Sign in	Have any feedback?	
Results										Copy as Markdown
Query #1 Execution time: 0.29ms										
emp_id	name	age	salary	department_id	hire_date					
1	John Doe	28	50000.0	1	2020-01-15					
2	Jane Smith	34	60000.0	2	2019-07-23					
3	Bob Brown	45	80000.0	1	2018-02-12					
4	Alice Blue	25	45000.0	3	2021-03-22					
5	Charlie P.	29	50000.0	2	2019-12-01					
6	David Green	38	70000.0	4	2022-05-18					
7	Eve Black	40	55000.0	3	2021-08-30					

2. Select only the name, salary columns from Employee Table.

Select name, salary from Employee;

Query #2 Execution time: 0.09ms	
name	salary
John Doe	50000.0
Jane Smith	60000.0
Bob Brown	80000.0
Alice Blue	45000.0
Charlie P.	50000.0
David Green	70000.0
Eve Black	55000.0

3. Select Employees who are older than 30

select name from Employee

where age > 30;

Query #3 Execution time: 0.17ms	
name	
Jane Smith	
Bob Brown	
David Green	
Eve Black	

4. Select names of all Departments

select name from Department;

Query #4 Execution time: 0.05ms

name
IT
HR
Finance
Marketing

5. Select Employess who work in the IT Department

select name from Employee where department_id=1;

(or)

SELECT e.name

FROM Employee e

JOIN Department d

ON e.department_id = d.department_id

WHERE d.name = 'IT';

Query #5 Execution time: 0.11ms

name
John Doe
Bob Brown

Query #6 Execution time: 0.11ms

name
John Doe
Bob Brown

6. Select Employees whose name start with 'J'

select name from Employee

where name like 'J%';

Query #7 **Execution time: 0.07ms**

name
John Doe
Jane Smith

7. Select Employees whose name end with 'e'

select name from Employee

where name like '%e';

Query #8 **Execution time: 0.06ms**

name
John Doe
Alice Blue

8. Select Employees whose name contains 'a'

select name from Employee

where name like '%a%';

Query #9 **Execution time: 0.07ms**

name
Jane Smith
Alice Blue
Charlie P.
David Green
Eve Black

9. Select employees whose names are exactly 9 characters long.

select name

from Employee

where name LIKE '_____';

Query #10 **Execution time: 0.06ms**

name
Bob Brown
Eve Black

10. 1select employees whose names have 'o' as the second character.

```
select name  
from Employee  
where name LIKE '_o%';
```

Query #11 Execution time: 0.06ms

name
John Doe
Bob Brown

11. Select employees hired in the year 2020.

```
select name  
from Employee  
where YEAR(hire_date)=2020;
```

Query #12 Execution time: 0.1ms

name	hire_date
John Doe	2020-01-15

12. Select employees hired in January of any year

```
select name  
from Employee  
Where MONTH(hire_date)=1;
```

Query #13 Execution time: 0.09ms

name	hire_date
John Doe	2020-01-15

13. Select employees hired before 2019.

```
select name  
from Employee  
where YEAR (hire_date) < 2019;
```

Query #14 Execution time: 0.07ms

name	hire_date
Bob Brown	2018-02-12

14. Select employees hired on or after March 1, 2021.

```
SELECT name,hire_date
```

```
FROM Employee
```

```
WHERE hire_date >= '2021-03-01';
```

Query #15 Execution time: 0.07ms

name	hire_date
Alice Blue	2021-03-22
David Green	2022-05-18
Eve Black	2021-08-30

15. Select employees hired in the last 2 years

```
select name,hire_date
```

```
from Employee
```

```
where
```

```
hire_date between '2020-01-1' and '2021-12-31';
```

Query #16 Execution time: 0.08ms

name	hire_date
John Doe	2020-01-15
Alice Blue	2021-03-22
Eve Black	2021-08-30

16. Select the total salary of all employees.

```
select sum(salary) as Total_salary
```

```
from Employee;
```

Query #17 Execution time: 0.07ms

Total_salary

410000.0

17. Select the average salary of employee.

select avg(salary) as avg_salary

from Employee;

Query #18 Execution time: 0.06ms

avg_salary

58571.428571

18. Select the minimum salary in the Employee table.

select name, salary

from Employee

where salary = (select min(salary) from Employee);

Query #19 Execution time: 0.13ms

name	salary
Alice Blue	45000.0

19. Select the number of employees in each department.

select department_id, count() as employee_count*

from Employee

group by department_id;

department_id	employee_count
1	2
2	2
3	2
4	1

20. Select the average salary of employees in each department.

```
select department_id, avg(salary) as avg_salary
from Employee
group by department_id;
```

department_id	avg_salary
1	65000.0
2	55000.0
3	50000.0
4	70000.0

21. Select the total salary for each department.

```
select d.name, SUM(e.salary) as total_salary
from Employee e
join Department d
on e.department_id = d.department_id
group by d.name;
```

Query #22 Execution time: 0.17ms

name	total_salary
Finance	100000.0
HR	110000.0
IT	130000.0
Marketing	70000.0

22. Select the average age of employees in each department.

```
select d.name, avg(e.age)
as avg_age
from Employee e
join Department d
on e.department_id=d.department_id
group by d.name;
```

Query #23 **Execution time: 0.14ms**

name	avg_age
Finance	32.5
HR	31.5
IT	36.5
Marketing	38.0

23. Select the number of employees hired in each year.

```
select YEAR(hire_date) as Year,count(*) as no_of_emps
from Employee e
group by YEAR(hire_date);
```

Query #24 **Execution time: 0.1ms**

Year	no_of_emps
2018	1
2019	2
2020	1
2021	2
2022	1

24. Select the highest salary in each department.

```
select d.name,MAx(salary)
from Employee e join Department d on d.department_id=e.department_id
group by d.name;
```

Query #25 **Execution time: 0.13ms**

name	MAx(salary)
Finance	55000.0
HR	60000.0
IT	80000.0
Marketing	70000.0

25. Select the department with the highest average salary.

```
select d.name, avg(e.salary) as avg_salary
from Employee e
join Department d
on e.department_id = d.department_id
group by d.name
```


Query #26 Execution time: 0.12ms

name	avg_salary
Finance	50000.0
HR	55000.0
IT	65000.0
Marketing	70000.0

26. Select departments with more than 2 employees.

```
SELECT department_id, COUNT(*) AS emp_count
```

```
FROM Employee
```

```
GROUP BY department_id
```

```
HAVING COUNT(*) > 2;
```

Query #27 Execution time: 0.08ms

department_id	emp_count
---------------	-----------

There are no results to be displayed.

27. Select departments with an average salary grester than 55000.

```
Select d.name,avg(salary) as avg_salary
```

```
from Employee e join Department d on e.department_id=d.department_id
```

```
group by d.name having avg(salary) > 55000;
```

Query #28 Execution time: 0.11ms

name	avg_salary
IT	65000.0
Marketing	70000.0

28. Select years with more than 1 employee hired.

```
SELECT YEAR(hire_date) AS Years, COUNT(*) AS emp_count
```

```
FROM Employee
```

```
GROUP BY YEAR(hire_date)
```

```
HAVING COUNT(*) > 1;
```

Query #29 Execution time: 0.12ms

Years	emp_count
2019	2
2021	2

29. Select departments with a total salary expense less than 100000.

```
SELECT d.name, SUM(e.salary) AS total_salary
FROM Employee e
JOIN Department d
ON e.department_id = d.department_id
GROUP BY d.name
HAVING SUM(e.salary) < 100000;
```

Query #30 Execution time: 0.13ms

name	total_salary
Marketing	70000.0

30. Select departments with the maximum salary 75000.

```
Select d.name, max(e.salary) AS max_salary
FROM Employee e
JOIN Department d
ON e.department_id = d.department_id
GROUP BY d.name
HAVING MAX(e.salary) = 75000;
```

Query #31 Execution time: 0.11ms

name	max_salary
There are no results to be displayed.	

31. Select all employees ordered by their salary in ascending order.

```
Select name,salary
from Employee order by salary;
```

Query #32 Execution time: 0.08ms

name	salary
Alice Blue	45000.0
John Doe	50000.0
Charlie P.	50000.0
Eve Black	55000.0
Jane Smith	60000.0
David Green	70000.0
Bob Brown	80000.0

32. . Select all employees ordered by their age in descending order.

select name,age

from Employee order by age desc;

Query #33 **Execution time: 0.05ms**

name	age
Bob Brown	45
Eve Black	40
David Green	38
Jane Smith	34
Charlie P.	29
John Doe	28
Alice Blue	25

33. Select all employees ordered by their hire date in ascending order

select name,hire_date

from Employee order by hire_date;

Query #34 **Execution time: 0.05ms**

name	hire_date
Bob Brown	2018-02-12
Jane Smith	2019-07-23
Charlie P.	2019-12-01
John Doe	2020-01-15
Alice Blue	2021-03-22
Eve Black	2021-08-30
David Green	2022-05-18

34. Select employees ordered by their department and then by their salary

select name,department_id,salary

from Employee order by department_id,salary;

Query #35 **Execution time: 0.19ms**

name	department_id	salary
John Doe	1	50000.0
Bob Brown	1	80000.0
Charlie P.	2	50000.0
Jane Smith	2	60000.0
Alice Blue	3	45000.0
Eve Black	3	55000.0
David Green	4	70000.0

35. Select departments ordered by the total salary of their employees.

```
select d.name, sum(e.salary) as total_salary
```

```
from employee e
```

```
join department d
```

```
on e.department_id = d.department_id
```

```
group by d.name
```

```
order by total_salary;
```

Query #36 Execution time: 0.2ms

name	total_salary
Marketing	70000.0
Finance	100000.0
HR	110000.0
IT	130000.0

36. Select employee names along with their departments names.

```
select e.name,d.name
```

```
from Employee e join Department d on e.department_id=d.department_id;
```

Query #37 Execution time: 0.12ms

name	name
John Doe	IT
Jane Smith	HR
Bob Brown	IT
Alice Blue	Finance
Charlie P.	HR
David Green	Marketing
Eve Black	Finance

37. Select project names along with the department names they belong to.

```
select p.name as pro_name, d.name as dept_name
```

```
from Project p join Department d on p.department_id=d.department_id;
```

Query #38 Execution time: 0.12ms

pro_name	dept_name
Project Alpha	IT
Project Gamma	IT
Project Beta	HR
Project Delta	Finance
Project Eta	Finance
Project Epsilon	Marketing
Project Zeta	Marketing

38. Select employees names and their corresponding project name

SELECT e.name, p.name AS project_name

FROM Employee e

JOIN Department d

ON e.department_id = d.department_id

JOIN Project p

ON d.department_id = p.department_id;

name	project_name
John Doe	Project Alpha
John Doe	Project Gamma
Jane Smith	Project Beta
Bob Brown	Project Alpha
Bob Brown	Project Gamma
Alice Blue	Project Delta
Alice Blue	Project Eta
Charlie P.	Project Beta
David Green	Project Epsilon
David Green	Project Zeta
Eve Black	Project Delta
Eve Black	Project Eta

39. Select all employees and their departments, including those without a department.

SELECT e.name, d.name AS department_name

FROM Employee e

LEFT JOIN Department d

ON e.department_id = d.department_id;

Query #40 **Execution time: 0.16ms**

name	department_name
John Doe	IT
Bob Brown	IT
Jane Smith	HR
Charlie P.	HR
Alice Blue	Finance
Eve Black	Finance
David Green	Marketing

40. Select all departments and their employees, including department without employees.

select d.name as dept_name,e.name as emp_name

from Department d left join Employee e on d.department_id = e.department_id;

Query #41 **Execution time: 0.07ms**

dept_name	emp_name
IT	John Doe
HR	Jane Smith
IT	Bob Brown
Finance	Alice Blue
HR	Charlie P.
Marketing	David Green
Finance	Eve Black